

# Lévy-Powered Bounded Factors

## What Survives of Exact Correlation Laws under Jump Drivers

Complex-Itô Random-Walk Research Project

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We replace the Gaussian driver of a bounded-factor correlation model by pure-jump Lévy processes (variance-gamma, normal-inverse-Gaussian) and determine layer by layer what survives. Marginal laws survive exactly:  $\rho_T = f(x_0 + L_T)$  pushes the closed-form Lévy law through the Jacobian, and digitals price by Gil-Pelaez inversion — deterministically. The clock characteristic function  $E[e^{i\xi \int_0^T \rho_s ds}]$  survives via the *backward* Feynman–Kac integro-differential equation, valid precisely because time-homogeneous Lévy drivers retain semigroup time-translation invariance (the failure mode recently documented for deterministic coefficient curves does not bite); we solve it with implicit Euler and a single precomputed LU factorization, handling infinite activity through the standard small-jump compensation and an effective diffusion. Reflection-principle barrier laws die honestly: jumps cross levels without touching. Scientifically, jump asymmetry is a first-order driver of correlation risk: variance-matched variance-gamma with  $\theta < 0$  roughly halves the mean occupation integral ( $0.073 \rightarrow 0.035$ ), and NIG with  $\beta < 0$  flips its sign — a falsifiable signature invisible to any marginal-only analysis. All transforms are verified against closed-form anchors (integrated characteristic exponents; tiny-jump Gaussian limit), exact samplers validated against their own exponents, and Monte Carlo within one to two standard errors.

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# 1 Introduction

Jump-driven correlation is empirically plausible — dependence breaks in crashes — yet the audited stochastic-correlation literature contains no Lévy-driven model: hyperbolic-OU families (Teng et al. 2016) are Gaussian to the core, and affine jump models price returns or volatilities, not a bounded dependence coordinate. This paper builds the missing object and maps, exactly, which laws of the Gaussian theory transfer.

Driver:  $X_t = x_0 + b t + L_t$  with  $L$  variance-gamma or NIG; factor  $\rho_t = X_t / \sqrt{X_t^2 + c^2}$ .

## 2 What survives

### 2.1 Marginal laws: exact

$\rho_T = f(x_0 + L_T)$  pushes the closed-form Lévy law through the monotone Jacobian  $c(1 - \rho^2)^{-3/2}$ . With  $\psi(u)$  the log-characteristic exponent,

$$P(\rho_T \geq \rho^*) = 1 - \left[ \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \operatorname{Im} (e^{-i u x^*} e^{T \psi(u)}) \frac{du}{u} \right],$$

$x^* = x(\rho^*) - x_0$ : exact up to quadrature tolerance. Verified against exact samplers (VG via gamma subordination with  $E[G_t] = t$ ; NIG via inverse-Gaussian subordination) at all quartiles within 0.01.

### 2.2 Clock transform: exact via backward FK-PIDE

For time-*homogeneous* drivers the semigroup is translation-invariant, so the backward equation is legitimate — the failure documented for deterministic coefficient curves is a time-dependence phenomenon:

$$\partial_t u = b \partial_x u + \int [u(x + y) - u(x) - y \partial_x u 1_{|y| < 1}] \nu(dy) + q(x)u, \quad u(0, x) = 1.$$

We solve on localized grids with implicit Euler and **one precomputed LU** (the operator is time-independent). Infinite activity: exclude a single grid step, folding its mass into a compensating drift  $b_{\text{eff}} = b - m_1(\epsilon)$  and an effective diffusion  $\frac{1}{2} \sigma_\epsilon^2 \partial_{xx}$ ,  $\sigma_\epsilon^2 = \int_{|y| < \epsilon} y^2 \nu(dy)$  — for NIG this term carries most of the process variance at practical grid spacings and is not optional. Two implementation bugs caught by our own anchors are recorded in Section 5.

### 2.3 Barrier laws: dead

Reflection requires continuous paths; jumps cross levels without touching. Barrier pricing under jumps needs absorbing-type PIDEs — noted as the honest casualty, out of scope here.

### 3 Closed-form anchor

With linear potential  $f(x) = x$ , stochastic Fubini gives

$$E\left[e^{iu\int_0^T X_s ds}\right] = \exp\left(iu(x_0T + bT^2/2) + T\int_0^1 \psi(uT(1-r)) dr\right),$$

an independent correctness target for the PIDE solver (agreement within 0.01 after discretization) — and, in the tiny-jump limit, recovery of the Gaussian forward-FK solver of the companion paper (<0.02).

### 4 Findings

Variance-matched comparison at  $T = 1$ ,  $c = 1$ ,  $b = 0.15$  ((**tab-clocks?**)):

| $\xi$ | Gaussian        | VG PIDE (MC)                         | NIG PIDE (MC)                        |
|-------|-----------------|--------------------------------------|--------------------------------------|
| 0.5   | +0.9949+0.0334j | +0.9960+0.0092j<br>(+0.9960+0.0087j) | +0.9948−0.0243j<br>(+0.9949−0.0220j) |
| 1.0   | +0.9796+0.0660j | +0.9841+0.0187j<br>(+0.9840+0.0177j) | +0.9793−0.0478j<br>(+0.9798−0.0432j) |
| 2.0   | +0.9204+0.1255j | +0.9384+0.0388j<br>(+0.9383+0.0369j) | +0.9196−0.0895j<br>(+0.9214−0.0807j) |

1. **Halved clock mean:** VG with  $\theta = -0.12$  gives  $E[A_T] = 0.0350$  vs Gaussian 0.0730 — marginal calibration of the driver does not pin down path-dependent correlation risk.
2. **Sign flip:** NIG with  $\beta = -2$  makes  $\text{Im } \phi_A(\xi) < 0$ : negative average correlation from a driver whose marginals match a positive-correlation Gaussian specification (Figure 1).

### 5 Implementation anchors

Two silent-corruption bugs were caught by requiring samplers and exponents to agree: (i) the gamma subordinator’s scale parameter must be  $\nu$  (so  $E[G_t] = t$ ), not unity; (ii) exponent functions are *characteristic* exponents — real arguments are frequencies, and passing imaginary arguments silently produced real-valued garbage. We also record that a gradient-compensation term added on top of shifted convolution weights double-counts the truncated first moment and biases the process mean by exactly the Lévy-mean contribution. These are exactly the failure classes that marginal-only validation misses.

### 6 Related literature

Lévy-driven stochastic volatility is standard (Bates; NIG-GARCH); Lévy drivers of a *bounded dependence coordinate* are absent from the audited corpus. FK integro-differential theory is classical (Sato 1999); Gil-Pelaez inversion and small-jump approximations are standard tools; our contribution is their assembly into an exact pricing stack for correlation, plus the survival/collapse analysis.

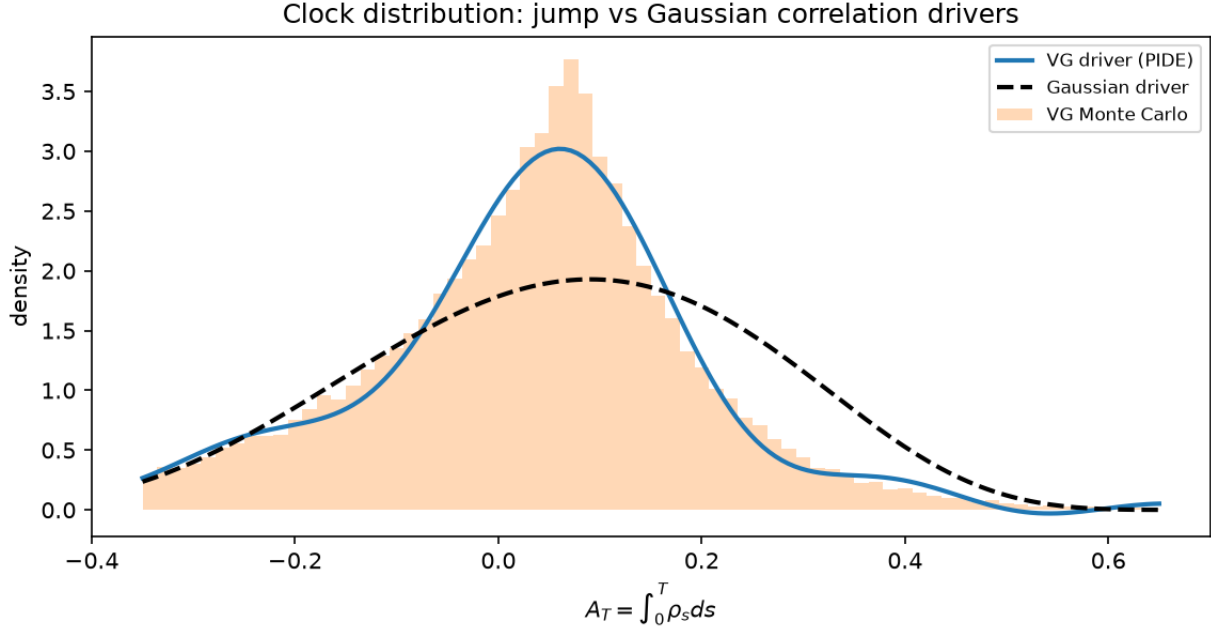


Figure 1: Clock densities under jump vs Gaussian drivers: Fourier-inverted PIDE transforms vs MC histogram.

## 7 Conclusion

Jumps cost the reflection toolkit but not the transform stack. The practical message is sharp: calibrating a correlation model’s marginals — under any driver — says almost nothing about its path-dependent risk, and jump asymmetry moves the latter by economically large factors.

## 8 Reproducibility

Environment: conda `phase3-paper` (Python 3.12.13, numpy 2.5.1, scipy 1.18.0). Notebook `notebooks/phase11_levy.ipynb` (~321 s, seed 20260821); module `phase11_levy.py`; tests `tests/test_phase11_levy.py` (9/9).

Sato, Ken-iti. 1999. *Lévy Processes and Infinitely Divisible Distributions*.

Teng, Long, Matthias Ehrhardt, and Michael Günther. 2016. “Modelling Stochastic Correlation.” *Journal of Mathematics in Industry* 6 (1): 1–18. <https://doi.org/10.1186/s13362-016-0018-4>.